Eclipses

LAB LESSON

OBJECTIVES

In this lesson, students will develop an understanding of solar and lunar eclipses. With the help of the video, they will investigate the differences between the eclipses and learn about the frequency of their occurences.

ACARA CONTENT DESCRIPTIONS

Predictable phenomena on Earth, including seasons and eclipses, are caused by the relative positions of the sun, Earth and the moon (ACSSU115)

- investigating natural phenomena such as lunar and solar eclipses
- modelling the relative movements of the Earth, sun and moon and how natural phenomena such as solar and lunar eclipses occur

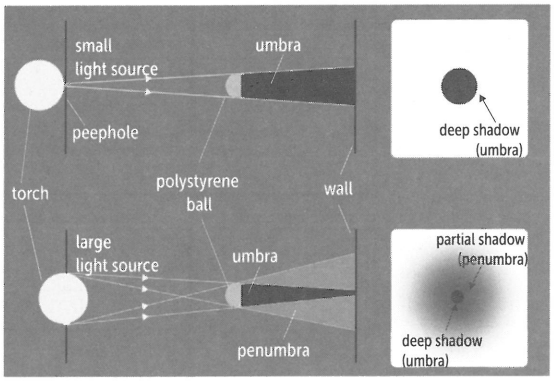
LESSON PLAN

Activities	Resources
Activity 1: The Legend of Skoll and Hati Open the presentation to the first slide and start the lesson with the mythical story of Skoll and Hati. Allow students to read the story on the slides. Lead a class discussion about why people from the past created myths to explain daily occurences (i.e. a lack of scientific knowledge and technological means to discover the truth). 10	• Presentation: Eclipses (first 9 slides)
Activity 2: The Two Types of Shadows Give out the The Two Types of Shadows worksheet to students. Using slides 11 - 14 of the presentation, discuss the difference between umbra and penumbra shadows and how they relate to the topic. For example: The **umbra** is the region of deep total shadow. When an umbra eclipse occurs, no sunlight is visible. The **penumbra** is the region of outer, partial shadow. When a penumbra eclipse occurs, some sunlight is visible. Follow the instructions below to demonstrate umbra and penumbra eclipses to the class. 1. Turn off the lights in the classroom. 2. Shine the torch against a blank wall. 3. Pierce the polystyrene ball with the skewer to create a lollipop-shaped figure. 4. Hold the figure about 1 m in front of the torch and approximately 1 m from the blank wall. This should create a penumbra shadow. 5. Create an umbra shadow, by holding the paper in front of the torch and shine the torch through the small opening. 20	• Presentation: Eclipses • Photocopies of the The Two Types of Shadows worksheet • Dark classroom • Blank wall, torch, polystyrene ball (approx. 10 cm in diameter), skewer, a piece of paper with a small opening cut out of the centre (the peeper)
Activity 3: I See You Hear Give out the I See You Hear worksheet and assign A or B to each pair of students: A - Watcher; B - Listener Play the *Solar Eclipses* video twice, (muting the audio). Only Student A will be allowed to watch the video. Student A will describe to Student B what was seen on the video, and together they will work together to come up with a script for the video with the help of the 'Cheat Section' and the diagrams on the worksheet. Repeat the same with the *Lunar Eclipses* video. After completing the task, ask students to read out their scripts to the class. Summarise the key points about eclipses with the rest of the presentation. 35	• Photocopies of the I See You Hear worksheet • ClickView video *Solar Eclipses* **https://clickv.ie./w/lp/11a** *Lunar Eclipses* **https://clickv.ie./w/lp/11b** • Presentation: Eclipses
Activity 4: All about Eclipses Give out the All about Eclipses worksheet and have students work on it individually to assess understanding. Use slides 11 and 12 of the presentation to review answers. 10	• Photocopies of the All about Eclipses worksheet • Presentation: Eclipses

ANSWERS

The Two Types of Shadows

1. Shadows are the result of light not being able to pass through objects.
2. The umbra is a region where no light can reach to while the penumbra is a region where some light can reach.
3. Suggested Answer:



I See You Hear

Possible Script:

Solar Eclipses

When the Moon passes between the Sun and Earth, the Sun temporarily disappears from the sky. The Sun, the Moon and Earth are aligned and this is a solar eclipse. The Moon can be seen blocking the Sun. When the Moon is close enough to Earth, its disc is sufficiently large enough to hide the Sun. From Earth, a total eclipse is observed. If the Moon is a little further away from Earth, an annular eclipse is observed where the Sun's outer edges remain visible to form a "ring of fire".

Lunar Eclipses

When the Earth is between the Sun and the Moon, the Moon is not lit by the Sun. An equivalent phenomenon causes lunar eclipses. In this case, Earth passes between the Sun and the Moon. The Moon is no longer lit up by the Sun. Instead, it turns dark red due to the deviation of the Sun's rays caused by Earth. This is known as a lunar eclipse.

All about Eclipses

Solar Eclipses

total solar eclipse, darkness, rare, new moon, smaller, close proximity, umbra

Lunar Eclipses

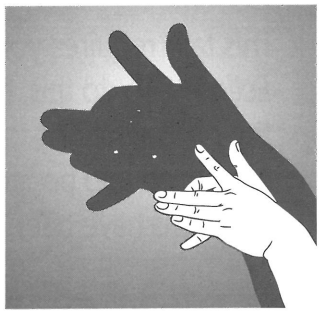
Earth's shadow, far side of Earth, full moon, common

EARTH AND SPACE SCIENCES

© Copyright ClickView Pty Limited 2016

The Two Types of Shadows

Answer the following questions.



1. Why do we see shadows?

_____

_____

_____

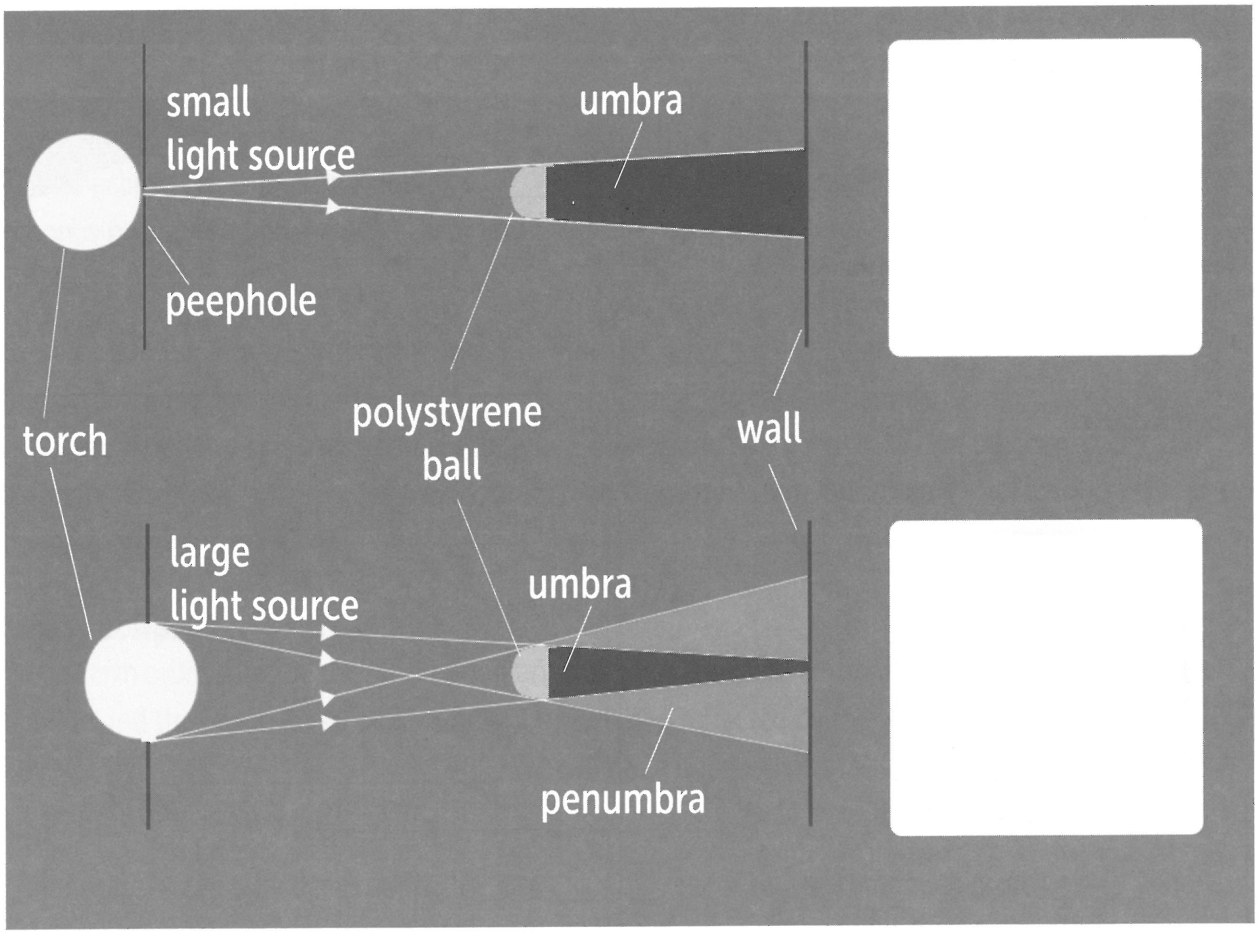
2. What is the difference between the umbra and penumbra regions of a shadow?

_____

_____

_____

3. As you watch the demonstration, draw and label a diagram to show what you observe on the wall.



© Copyright ClickView Pty Limited 2016

I See You Hear

Instructions:

This task requires you to work in pairs to create new scripts for two videos - *Solar Eclipses* and *Lunar Eclipses*. To do this, the video will be played on mute. Student A will watch the video and relay their observations to Student B, who will be facing away from the screen. Student B should have a notepad and pen to take detailed notes. The videos will be played twice. A 'Cheat Section' and diagrams have been provided to help you create your scripts. Good luck!

Cheat Section

The Sun is the source of light for both types of eclipses.

Umbra – the innermost and darkest region of a shadow

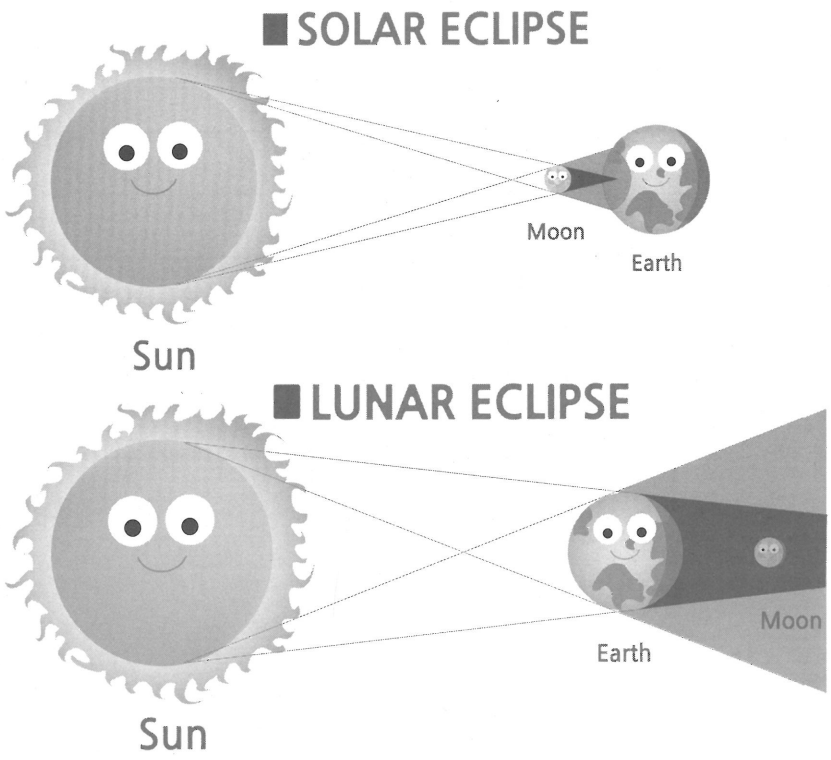
Penumbra – the outer, partial region of a shadow

Solar eclipse - when the moon blocks the light of the sun from reaching Earth

Lunar eclipse - when Earth blocks the sunlight that normally is reflected by the moon

Annular eclipse - a type of solar eclipse, when the moon is farthest from Earth and does not block the entire view of the sun, creating what looks like a ring around the moon.

Total eclipse - an eclipse in which the whole of the disc of the Sun or Moon is obscured



Solar Eclipse	Lunar Eclipse

© Copyright ClickView Pty Limited 2016

All about Eclipses

How much do you know about eclipses? Fill in the blank spaces with words or phrases from the box below.



Earth's shadow	smaller	rare	darkness	full moon	new moon
total solar eclipse	close proximity	umbra	common	further	far side of Earth

Solar Eclipses

When the Sun, Moon and Earth are perfectly aligned, the Moon blocks the light of the Sun from reaching Earth. This is called a _____. On Earth we experience temporary _____caused by a big, round shadow (the Moon) blocking our sunlight. Total solar eclipses are _____ and scientists believe that, in the very best conditions, they only last for around seven minutes.

Partial solar eclipses occur when the Moon passes between the Sun and Earth but they are not perfectly aligned, allowing only part of the Sun to be covered. Partial solar eclipses are much more common than total solar eclipses.

A solar eclipse can only occur during a _____, a phase of the lunar cycle where the Moon is positioned between the Earth and the Sun. During a new moon phase, the Moon is barely visible from Earth.

Even though the Moon is approximately 400 times _____ than the Sun, it appears to be the perfect size to cover the Sun due to its _____ to Earth. Whether you see a full or partial eclipse will depend on where you are located in the world. A total eclipse can be observed if your location falls in the _____ of the Moon. A partial eclipse can be observed if your location falls in the penumbra of the Moon.

Lunar Eclipses

While Earth orbits the Sun, the Moon simultaneously orbits around Earth. When Earth passes between the Sun and the Moon it can block sunlight from being reflected by the Moon. Instead, the Moon is covered by _____. We call this a lunar eclipse. A lunar eclipse can only occur when the Moon is on the _____, away from the Sun.

Lunar eclipses can only occur during a _____. Lunar eclipses are much more _____than solar eclipses. They are also easier for many people to view at the same time. Scientists believe it is possible for a whole hemisphere to see a lunar eclipse occurring. Lunar eclipses can last several hours.

© Copyright ClickView Pty Limited 2016